

Farsi Review Rating Prediction Using Prompt Engineering and RAG

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ABSTRACT

In this study, we propose several prompt-based methods for predicting ratings of reviews using large language models, specifically GPT-3.5 Turbo and Dorna. We utilize BaSalam dataset, sourced from an Iranian online marketplace. Our approach includes zero-shot and few-shot prompting, as well as Retrieval-Augmented Generation.

We evaluate the effectiveness of our methods by comparing them to baseline models, demonstrating superior performance in terms of mean absolute error (MAE) and mean squared error (MSE).

After conducting several experiment, the results shows the superiority of in-context learning method with examples retrieved.

INTRODUCTION

This study explores methods for predicting ratings for review texts, a task similar to sentiment analysis and due to the inherent ordinal nature of ratings framed as a regression problem predicting ratings from 1 to 5.

Given our focus on prompting techniques, we utilized two large language models: GPT-3.5 Turbo and Dorna. GPT-3.5 Turbo, a proprietary model from OpenAI, and Dorna, an open-source Farsi language model known for its strong performance on various benchmarks, both play a key role in our study. It is worth noting that Dorna is a family of decoder-only models, specifically trained and fine-tuned on Farsi data, developed by Part AI. We use an 8B instruct model from this family.

Finally, the dataset comprises review texts sourced from BaSalam, an Iranian online platform.

METHODOLOGY

The preprocessing steps include removing empty reviews and duplicates and extracting a balanced subset for analysis using under sampling. Our methods are:

1. Baseline Methods

As baseline methods, we use SVM and KNN, using multilingual E5 base embeddings as their features.

2. Zero-Shot Prompting

We employ zero-shot prompting, where the model is not provided with any examples. Instead, it is tasked solely with predicting the rating for a specific review.

3. Few-Shot Prompting

In this method, we utilize two distinct types of few-shot prompting. The first approach involves providing the model with fixed examples of each class and the second approach focuses on the user's past reviews; specifically, we supply the model with examples of each available rating class, chosen randomly from the same user's previous reviews.

4. Retrieval-Augmented Generation(RAG)

In this method, we retrieve the most similar examples to the target review for which we aim to predict the rating. For the retrieval process, we utilize the multilingual E5 base to generate embeddings. These examples are incorporated into the prompt to provide relevant context for predicting the rating through in-context learning.

An example of our prompt:

As a large language model your task is to predict user ratings for products on an online Iranian marketplace based on the provided product review.

Since the reviews are in Persian, be sure to leverage your capabilities for understanding nuances and sentiment within the Persian language. This will improve the accuracy of your predictions. You will be given a set of similar examples containing reviews and their corresponding ratings (1-5) for better understanding.

Examples:

Review: {Review Text 1} , Rating: {Rating Score 1}

Review: {Review Text 2} , Rating: {Rating Score 2}

...

Review: {Review Text n} , Rating: {Rating Score n}

Complete this prompt only with one number (1, 2, 3, 4 or 5) as your prediction. Further explanations are not needed.

Review: "" {Target Review} "", Rating:

RESULTS

We conducted multiple experiments to predict review ratings using our dataset. We employed RAG with two, four, and eight examples. Moreover, we explored two methods for few-shot prompting, one with fixed examples and one with user-specific examples. Our key findings are:

- Performance consistently improves with the use of more examples in RAG setting.
- RAG with eight examples have the lowest MAE among all methods.
- Few-shot prompting with user-specific examples yields better results than with fixed examples. This improvement is attributed to the unique rating behaviors of individuals.
- Few-shot with user-specific examples achieves the lowest MSE among all methods.

Table 1: Comparison of MAE and MSE for various prediction methods.

Models	Methods	MAE	MSE
Baselines	SVM	0.790	1.110
	KNN	<u>0.760</u>	<u>1.048</u>
GPT-3.5	Zero-Shot	0.632	0.924
	Fixed-Example	0.603	0.851
	User-Specific Examples	0.602	<u>0.834</u>
	RAG with Two Examples	0.601	0.893
	RAG with Four Examples	0.589	0.867
	RAG with Eight Examples	<u>0.580</u>	0.864
Dorna	Zero-Shot	0.729	0.943
	Fixed-Example	0.721	1.061
	User-Specific Examples	0.679	1.037
	RAG with Two Examples	0.727	1.145
	RAG with Four Examples	0.649	0.995
	RAG with Eight Examples	<u>0.614</u>	<u>0.870</u>

We also conducted another experiment to test GPT-3.5 in two language settings.

1. **Farsi:** The entire prompt is in Farsi.
2. **Mixed-language:** The task description is in English, but the examples are in Farsi.

Results indicate that when the entire prompt is in Farsi, the performance is superior.

Table 2: Comparison of GPT-3.5 with different language settings.

Language	Methods	MAE	MSE
Farsi	Zero-Shot	0.632	0.924
	User-Specific Examples	0.602	<u>0.834</u>
	RAG with Eight Examples	<u>0.580</u>	0.864
Mixed Setting	Zero-Shot	0.651	1.053
	User-Specific Examples	0.643	<u>0.929</u>
	RAG with Eight Examples	<u>0.611</u>	0.949